Unit 1

Graphics Primitives & LINE GENERATION

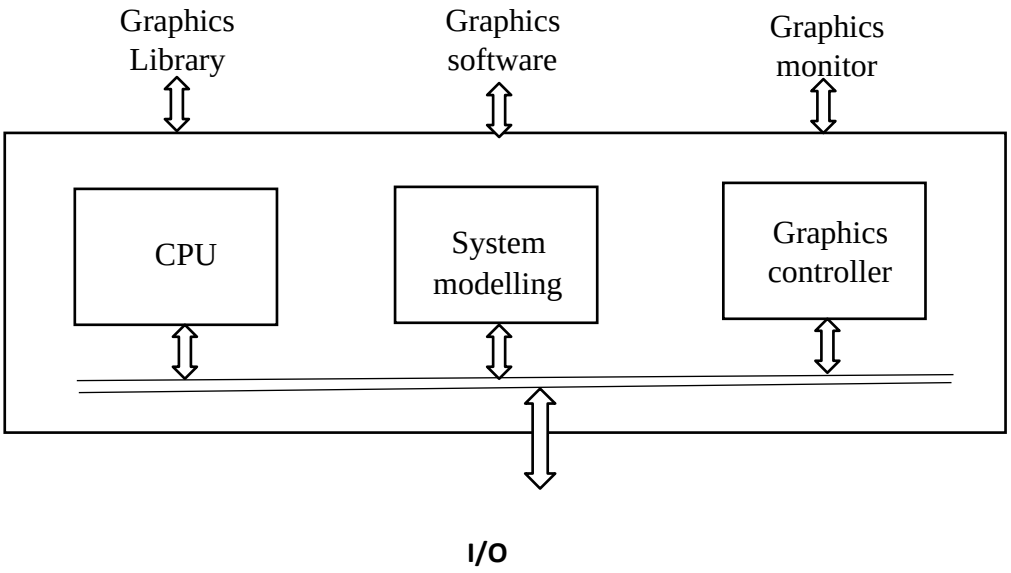
computer Graphics

Computer graphics is the art of drawing images, line segments, graphs, as well as other objects using computers and programming. A computer graphics image is form with pixels that is the smallest addressable unit represented on a computer screen is the pixel.

Points to Remember

- Computer is information processing machine. User needs to communicate with computer and the computer graphics is one of the most effective and commonly used ways of communication with the user.
- It displays the information in the form of graphical objects such as pictures, charts, diagram and graphs.
- Graphical objects convey more information in less time and easily understandable formats for examplestatically graph shown in stock exchange.
- In computer graphics picture or graphics objects are presented as a collection of discrete pixels.
- We can control intensity and color of pixel which decide how picture look like.
- The special procedure determines which pixel will provide the best approximation to the desired pictureor graphics object this process is known as **Rasterization.**
- The process of representing continuous picture or graphics object as a collection of discrete pixels iscalled **Scan Conversion.**

Graphics Hardware System



I/O devices

Advantages of computer graphics

- Computer graphics is one of the most effective and commonly used ways of communication withcomputer.
- It provides tools for producing picture of "real-world" as well as synthetic objects such as mathematical
  surfaces in 4D and of data that have no inherent geometry such as survey result.
- It has ability to show moving pictures thus possible to produce animations with computer graphics.
- With the use of computer graphics we can control the animation by adjusting the speed, portion ofpicture in view the amount of detail shown and so on.
- It provides tools called motion dynamics. In which user can move objects as well as observes as perrequirement for example walk throw made by builder to show flat interior and surrounding.
- It provides facility called update dynamics. With this we can change the shape color and other propertiesof object.
- Now in recent development of digital signal processing and audio synthesis chip the interactive graphicscan now provide audio feedback along with the graphical feed backs.

Application of computer graphics

- User interface: - Visual object which we observe on screen which communicates with user is one of the most useful applications of the computer graphics.
- Plotting of graphics and chart in industry, business, government and educational organizations drawing like bars, pie-charts, histogram's are very useful for quick and good decision making.
- Office automation and desktop publishing: - It is used for creation and dissemination of information. It is used in in-house creation and printing of documents which contains text, tables, graphs and other forms of drawn or scanned images or picture.
- Computer aided drafting and design: - It uses graphics to design components and system such as automobile bodies structures of building etc.
- Simulation and animation: - Use of graphics in simulation makes mathematic models and mechanicalsystems more realistic and easy to study.
- Art and commerce: - There are many tools provided by graphics which allows used to make their pictureanimated and attracted which are used in advertising.
- Process control: - Now a day's automation is used which is graphically displayed on the screen.
- Cartography: - Computer graphics is also used to represent geographic maps, weather maps,oceanographic charts etc.
- Education and training: - Computer graphics can be used to generate models of physical, financial andeconomic systems. These models can be used as educational aids.
- Image processing: - It is used to process image by changing property of the image.